

JOHNATHAN AHDOUT

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EDUCATION

Princeton University

B.A. Mathematics; Minors: Computer Science & Optimisation and Quantitative Decision Science

Princeton, NJ
Present – May 2029

• GPA: 3.79/4.00

• Relevant Coursework: *Honours Linear Algebra, Single-Variable Real Analysis, Probability Theory, Data Structures and Algorithms, Abstract Algebra, Combinatorics, Financial Investments & Derivatives, Multivariable Calculus, Fundamentals of Statistics*

John L. Miller Great Neck North High School

High School Diploma

Great Neck, NY
Sep 2021 – Jun 2025

• Salutatorian; GPA: 97.8/100; SAT: 1590 (800 Math, 790 Reading & Writing)

RESEARCH & EXPERIENCE

Dickson Lab @ Michigan State University, ACRES REU

NSF-Funded Research Intern — Generative AI for Molecular Design

East Lansing, MI
May 2026 – Present

- Selected as 1 of 10 students from 200+ applicants for the NSF-funded ACRES REU to conduct ML research under Dr. Alex Dickson.
- Extended the LEVI molecular generation framework in PyTorch by developing SE(3)-equivariant graph neural network & diffusion-model methods for generating chemically valid drug candidates inside protein binding pockets.
- Improved distributed multi-GPU training & inference on Michigan State's high-performance computing cluster while evaluating new diffusion-model architectures for molecular inpainting & protein-conditioned generation.
- Developed **ACForce**, a novel graph-connectivity objective based on Laplacian graph theory that increased molecular connectivity by over **30%** & by nearly **50%** on benchmark drug-like molecule datasets while preserving molecular quality. Continuing work to implement network flow & effective resistance methods to prevent overcompacted & valence-inconsistent molecule generation.
- Presented research at the MSU Mid-SURE Conference ([poster](#)), earned **1st Place Presentation** at the ACRES Symposium, produced rigorous theoretical analysis of the proposed method, and preparing manuscripts to publish as the **first-author of one paper** and the **second-author of another** currently being prepared for submission to NeurIPS and ICLR.

Independent Research, Princeton University

Conditional Mutual Information for Point-in-Time Macro-Quantamental Screening

Princeton, NJ
Aug 2026 – Present

- Developed a [research proposal](#) applying conditional mutual information to screen point-in-time macroeconomic indicators from J.P. Morgan's Macrosynergy Quantamental System (JPMaQS) for predictive content beyond linear & existing information-theoretic measures. Working with Macrosynergy Quant Researchers & Princeton ORFE faculty to develop and publish.
- Derived the bias-variance rate of a k -nearest-neighbour conditional-mutual-information estimator & designed a cross-country panel-pooling procedure, with a poolability test, to relax its dimensionality limit.
- Proposed a diffusion-model-based estimator, built on recent score-based mutual-information methods, to admit high-dimensional text embeddings (FOMC statements, earnings-call transcripts) as screening indicators.

EXTRACURRICULAR INVOLVEMENT & PROJECTS

Princeton Puzzles Club

Officer & Princeton Puzzlehunt Puzzle Writer/Tester

Sep 2025 – Present

- Design & test competition-grade logic puzzles by constructing multi-step solution paths, verifying uniqueness, balancing solve difficulty, and identifying edge cases.
- Helped design and test three puzzles for Princeton Puzzlehunt 2026, supporting a competition with 500+ participants through live debugging, grading, & real-time hint management.

Options Pricing & Volatility-Trading Research Toolkit (Python)

Jun-Aug 2026

- Engineered three pricing engines (Black-Scholes, Monte Carlo, CRR binomial tree) behind a shared interface.
- Benchmarked Newton-Raphson, Brent, and Jaeckel/Halley IV solvers across 250 cases; Newton diverges on **14.4%** of a realistic grid, motivating Jaeckel as the default.
- Derived a closed-form gamma/theta P&L identity from Itô's lemma and the Black-Scholes PDE, validated against a simulated self-financing delta-hedge.
- Implemented a 2-state Gaussian HMM from scratch (Baum-Welch EM, Viterbi) to detect volatility regimes and drive a gamma-scalping strategy.
- Built a 97-test suite (pytest) with regressions for bugs found during development, and generated a live SPY implied-vol surface exposing the smile absent from constant-volatility Black-Scholes.

GridSmart: Real-Time Grid Optimisation & Smart Home Scheduling

Apr 2026

Princeton University Claude Builder Club Hackathon

- Built an AI-powered electricity optimisation platform integrating ERCOT pricing, SmartThings, and Alexa APIs to automatically schedule household energy usage based on cost, renewable availability, and grid demand.
- Designed an LLM-powered data ingestion pipeline and interactive dashboard that optimised electricity costs while analysing regional energy equity and renewable utilisation.
- Presented the [project](#) to Princeton ORFE and Computer Science faculty during the final judging round.

SKILLS, AWARDS & DISTINCTIONS

Languages, Tools, and Proficiencies: Python, Java, R, PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, L^AT_EX, Excel, PowerPoint

Certification: [J.P. Morgan Quantitative Research Job Simulation](#) (Forage)

Honours:

- National Merit \$2500 Scholarship Recipient (2025)
- U.S. Presidential Scholar Semifinalist (2025) | Coca-Cola Scholarship Semifinalist (2025) | Hagan Scholarship Finalist (2025)
- **Regeneron STS 2025 Top 300 Scholar:** "[Contextualising Data-Sparse QSPR Property Prediction with Categorical Representations Built from Variational Autoencoders](#)" (accepted to Princeton Undergraduate Research Journal)
- Co-author, *PLOS Biology*: "[Developing Inhibitory Peptides Against SARS-CoV-2 Envelope Protein](#)"
- Quiz Bowl NAQT Nationals Qualifier (2024, 2025); Princeton Quiz Bowl A-Team (Top 10, ACF Winter Regionals)
- 2× IAC National Science Bee Nationals Qualifier (2024, 2025), national placements of 15th and 4th.